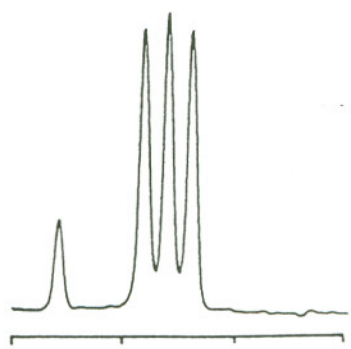
PHYSICAL CHEMISTRY CUMULATIVE EXAM

February 12, 1994

Table of isotopes

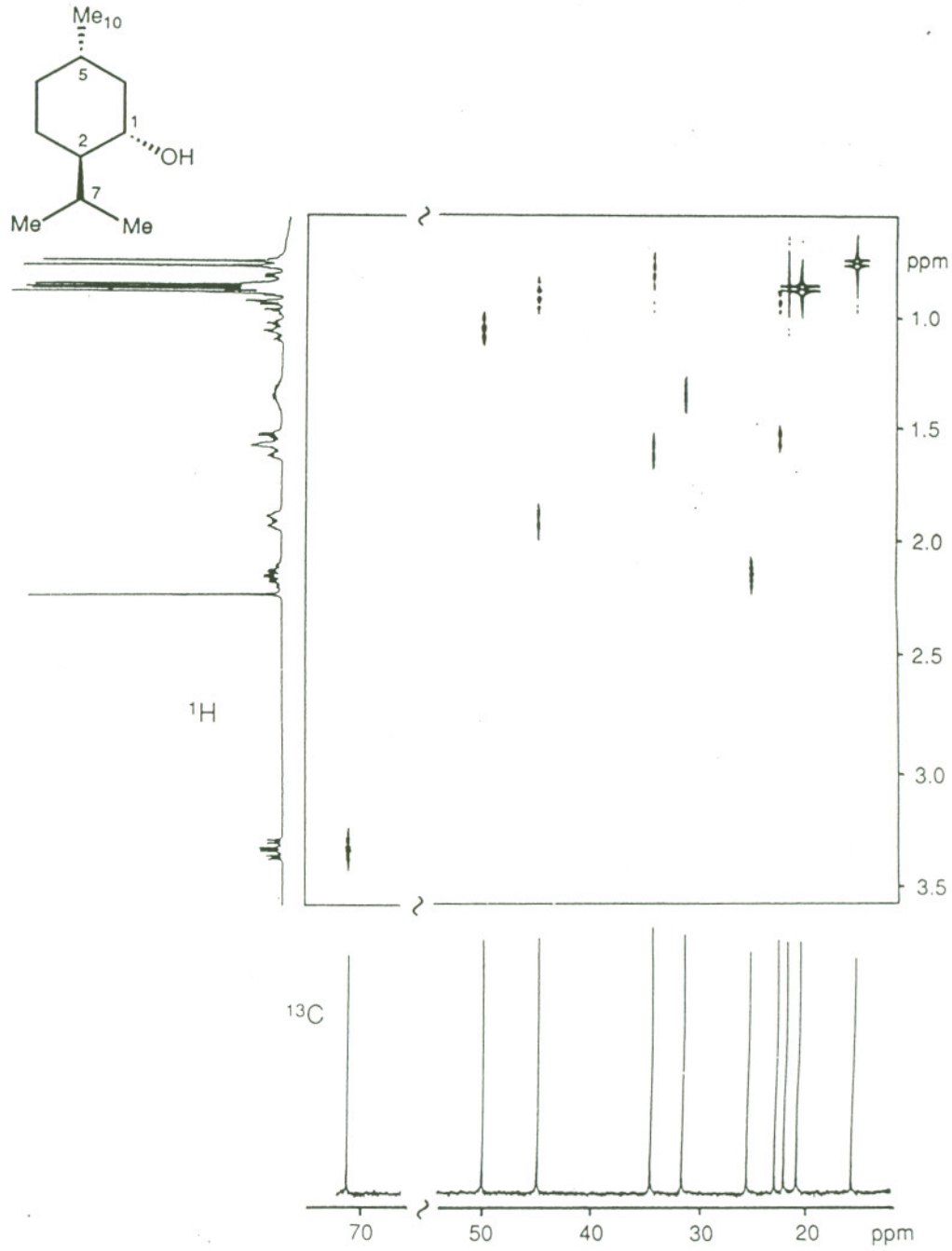
Nucleus	I	Nat. abundance (%)	Rel. frequency (MHz)
^{1}H	1/2	99.98	100.00
^{2}H	1	0.016	15.35
^{6}Li	1	7.42	14.71
^{10}B	3	19.58	10.75
^{11}B	3/2	80.42	32.08
^{13}C	1/2	1.11	25.14
^{14}N	1	99.63	7.22
^{15}N	1/2	0.37	10.13
^{19}F	1/2	100.0	94.08
^{31}P	1/2	100.0	40.48
^{47}Ti	5/2	7.28	5.637
^{49}Ti	7/2	5.51	5.638
^{51}V	7/2	99.76	26.29
^{119}Sn	1/2	8.58	37.27
^{181}Ta	7/2	99.98	11.97
^{187}Os	1/2	1.64	2.30
^{195}Pt	1/2	33.8	21.50
^{199}Hg	1/2	16.84	17.83

1. The 400 MHz ^{1}H spectrum of a sample labelled 'CD_2Cl_2' is shown here. Assign and explain the signals, which have a chemical shift of around 5.2 ppm. The scale markers are 5 Hz each.

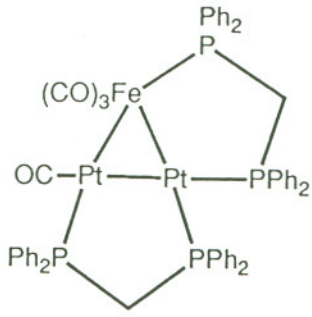


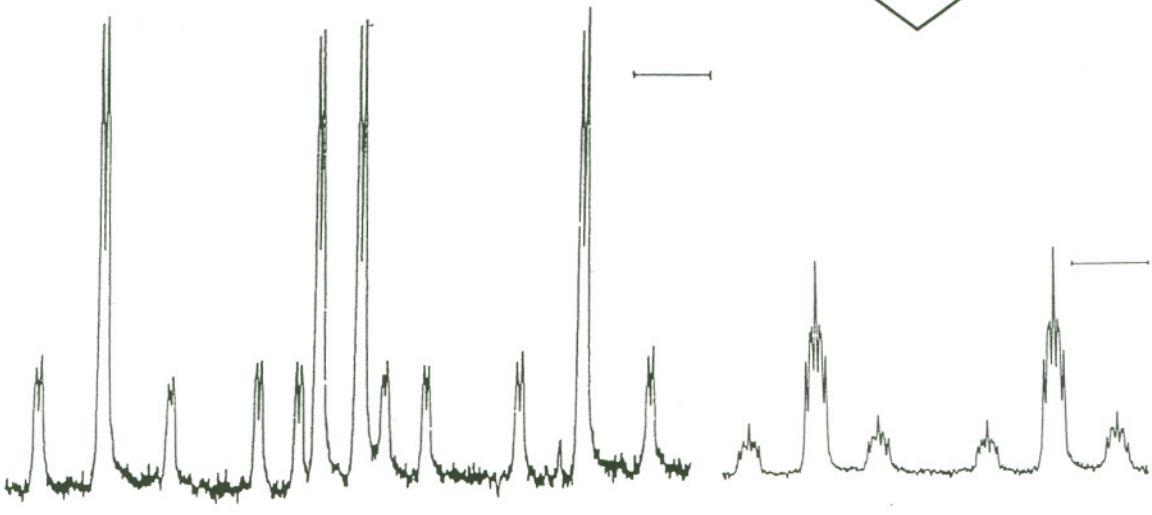
2 Assign as much as possible of the ^{1}H and ^{13}C spectra of menthol using the ^{13}C INADEQUATE spectrum below and the one-bond C–H correlation opposite.

Conditions: $CDCl_3$ solution, 300 MHz ^{1}H and 75 MHz ^{13}C spectra. The spectrum below was obtained with a modified pulse sequence that gives a symmetrical spectrum. The cross-peaks contain doublets because the spectrum is detecting coupled ^{13}C–^{13}C pairs.

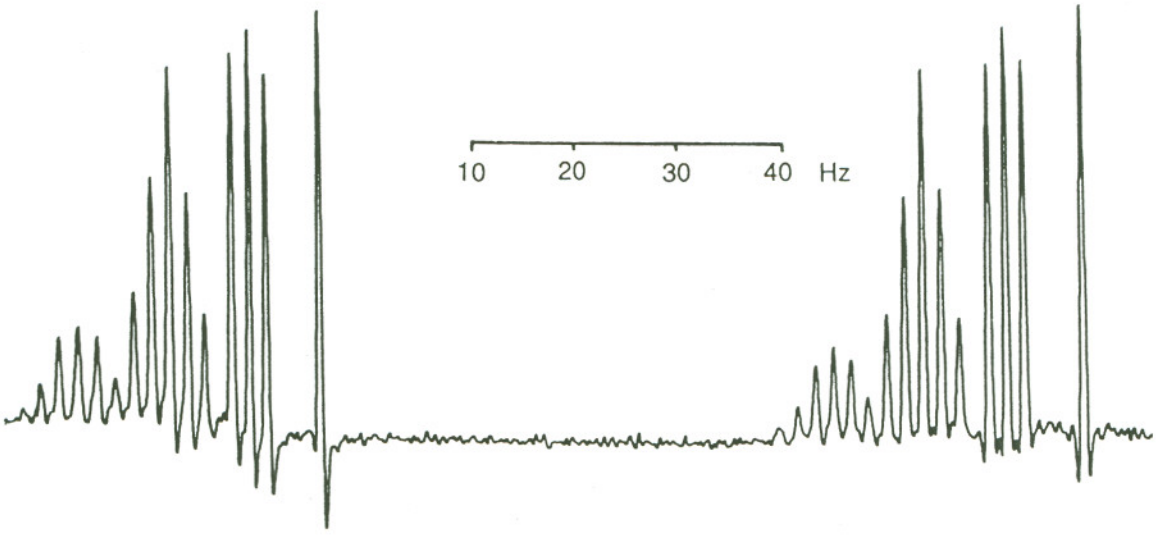


3 Assign and interpret the major features of the two multiplets from the ^{1}H-decoupled ^{195}Pt spectrum of this complex. The bar represents 1000 Hz for both multiplets.



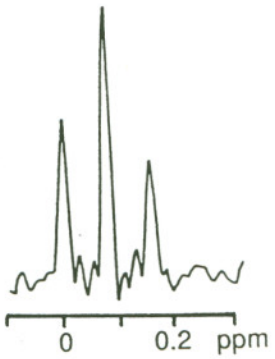
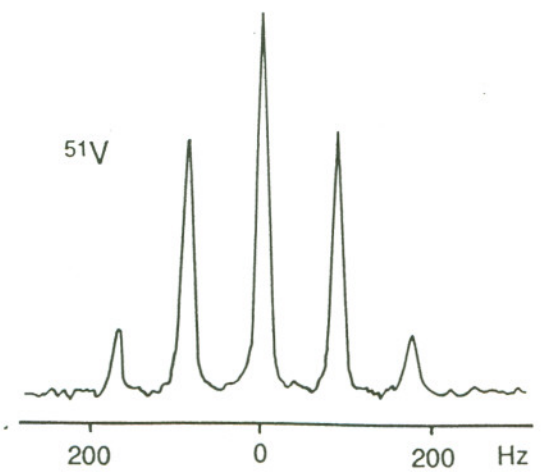
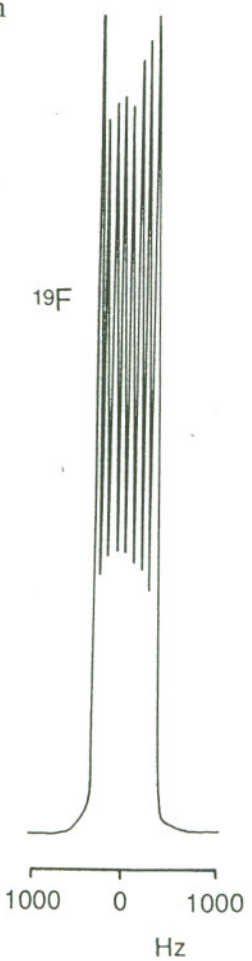


4 ^{15}N-Labelled ammonium chloride (NH_4Cl) was dissolved in a mixture of H_2O and D_2O and then strongly acidified. Interpret as fully as possible the resulting ^{1}H spectrum shown below. Conditions: 400 MHz; only the NH region is shown.



5 The bacterium *Nitromonas* biosynthesizes 3-nitropropanoic acid, **A**, from ammonia, NH_3. If the bacterium is grown on ^{15}N-labelled ammonia and natural-abundance oxygen gas ($^{16}O_2$), the ^{1}H-decoupled ^{15}N resonance of **A** is a sharp singlet; the shift of this singlet is arbitrarily defined as zero. When the experiment is repeated with ^{15}N-labelled ammonia and a 1:1 mixture of $^{18}O_2$ and $^{16}O_2$, **A** has the ^{15}N spectrum shown. What conclusions can be drawn about the source and mechanism of oxygen incorporation?

A



6 The ^{51}V and ^{19}F spectra of the complex ion $[VOF_4]^-$ are shown here. What is the structure of the ion?